Structure optimization of Cu and Adsorption analysis of H_2O and NH_3 on the Cu(111) surface

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1 Introduction

Surface adsorption plays an important role in many catalytic processes. Moreover, the surface of the catalyst is exposed to ambient conditions, where interaction with water is one of the most important aspects of the process and may substantially influence the properties of the catalyst. Here we study indipendently the molecular adsorption (physisorption) and dissociative adsorption (chemisorption) of H_2O and NH_3 on an ideal Cu(111) surface, including also Van der Waals corrections. The surface is constructed from scratch, starting from the bulk structural optimization, and then constructing a multi-layer slab.

2 Methods

We have carried out electronic structure calculations using DFT as implemented in Quantum ESPRESSO [1]. The total energy calculations have been performed using the Perdew – Burke – Ernzerhof (PBE) [2] form of the generalized gradient approximation (GGA). The upper values of cut-off energies for the plane wave expansion and the number of k-points in the Monkhorst–Pack grid are obtained during the structural optimization of the bulk of Cu, converging every single value of interest indipendently. Efficiency ultra-soft pseudopotentials [3] are used for every atomic species, for these kind of pseudos the recommended cut-off for the density is eight times the energy cut-off. The pseudopotential implements non-linear Core Correction and a scalar relativistic treatment of the spin-orbit coupling. The suggested minimum energy cut-off is between 44 and 47 Ry. To improve the description of the long-range interaction, we have considered the DFT-D2 Grimme's [4] correction of the Van der Waals forces, primarely to better treat physisorption of polar molecules, such as H_2O and NH_3. The k-points shift from the Γ-point is chosen so that the number of inequivalent k-points is minimum. While in the fcc cubic cell of bulk Cu the shift reduces k-points, the opposite is true for the Cu(111) slab, inside an exagonal cell [5]. Since we are dealing with a metal, a gaussian smearing with 0.02 Ry smearing is introduced in each calculation.

3 Results and discussion

The procedure followed in this calculation is divided in three steps:

- Bulk optimization: we have converged the energy cutoff E_{cut} and the number of k-points n_k for a $n_k \times n_k \times n_k$ grid, then we have calculated the atomic distance, bulk modulus and cohesive energy of copper.
- Surface optimization: we have relaxed the multi-layer slab structure to find the surface energies varying slab thickness. Another convergence is helpful to determine the minimum safe vacuum size for the adsorption step. The relaxed structure is used as a starting point for adsorption calculations.
- Adsorption analysis: the molecule or a molecular fragment is placed inside the cell. A

lot of possible high symmetry starting configuration are probed; the physisorbed state is obtained positioning the molecule in the most stable site, while in the chemisorbed state the fragments are placed in a 2×2 supercell in their most stable configuration (when possible). These relaxed states can be placed as the starting and final configuration for a Nudged Elastic Band (NEB) calculation. From the results we can directly estimate the energy barrier across the minimum energy reaction path in both directions and verify that there are no other candidates for the most stable physisorbed and chemisorbed state. Finally an interface is constructed and the reaction is analysed with various loads.

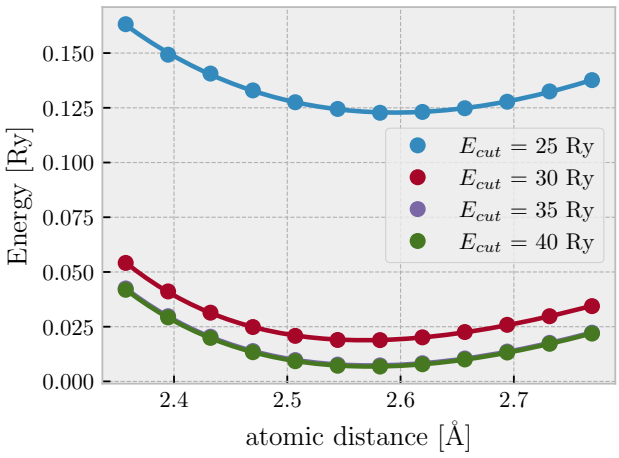


Figure 1: Convergence of E_{cut}, the energy is shifted arbitrarily. $E_{cut} = 35$ Ry data coincide quite exactly with $E_{cut} = 40$ Ry. Points are taken from run with different (fixed) lattice parameter and fitted with the Murnaghan equation of state.

3.1 Bulk properties

The convergence of E_{cut} is summarized in Fig. (1): the cell parameters are relaxed manually, performing a SCF for some values of the atomic distance a near the equilibrium distance a_0; then E_{cut} is changed and the same cycle is repeated. From the figure it's clear that for cut-off higher than 35 Ry the bulk system is the same. The resulting points are fitted with a suitable equation of solid state, such as the Murnaghan equation [6], the equilibrium distance a_0 is calculated as the minimum of the energy curve, the bulk modulus b_0 as the convexity of the curve. The results are summarized in Tab. (1). Since we are not interested in precision calculations and since these equilibrium values do not change more than a few percent varying the cut-off, we have chosen $E_{cut} = 30$ Ry for the following calculations. We have then calculated the energy of the isolated copper atom, placed in a cubic supercell of 20 Å; the cohesion energy is estimated as the energy difference between crystal and isolated state. The convergence on the number of k-points is conducted in a similar fashion: from Fig. (2) we can notice that for n_k higher than four the error introduced is no more than 10^{-2} eV.

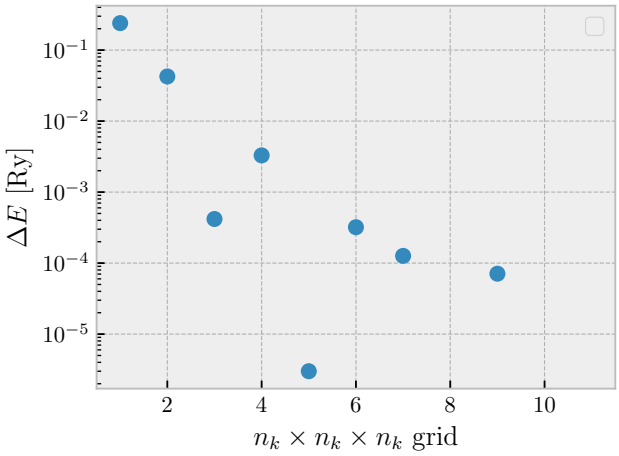


Figure 2: Convergence of k-points, ΔE is referred to the energy of the $n_k = 10$ grid.

E_{cut} (Ry)	a (Å)	b_0 (GPa)	E_{coh} (eV/at)
25	2.595	149	1.94
30	2.568	140	3.36
35	2.574	136	3.51
40	2.573	137	3.52
lit.	2.561	145	3.49

Table 1: Comparison between literature [7] and calculated equilibrium bulk values for the energy cut-offs taken in analysis. The cohesive energy is compared with an experimental value [8].

3.2 Surface properties

The Cu(111) surface is modelled from a 2×2 hexagonal supercell replicated in the xy plane, this precaution is taken to be sure that the molecule's fragments are all fitted inside the unit cell. The two parameters of interest are the number of layers of the slab and the vacuum size. The k-points grid is $3 \times 3 \times 1$ with no shift. Surface energy is obtained from:

$$E_{surf} = (E_{slab} - NE_{bulk})/2A,$$

where N is the number of Cu atoms in the slab and A the unit cell's area in th xy plane. Slab energy is obtained performing a BFGS relaxation on the ideal slab. Starting from 10 Å of vacuum we have tested various number of layers (from 3 to 12): resulting surface energies do not vary more than few percent from each other, with the maximum surface energy of 1.23 J/m^2 obtained with 3 layers, to be compared with literature computed value of 1.34 J/m^2 [7]. Adsorption calculations are performed with a 3 layers slab. Performing convergence on the vacuum size, we have seen that it can be reduced until 5 Å with an energy difference of no more than 0.1 eV; this result demonstrate that the electron cloud inside the slab tails out very steeply, but adding adsorbates to the structure can represent a problem for such low vacuum thicknesses, hence we have considered 10 Å as a conservative choice of this parameter. We are not interested in other surfaces since (111) is the (only) close packing direction of cubic fcc structures. Coordinates of the relaxed system is then exported for all the adsorption calculations.

3.3 Adsorption properties

It is not possible to probe the entire configuration space of the positions and orientation of the adsorbed species, then a faster strategy is needed. Four high-symmetry adsorption sites are identified for the fcc surface: one "ontop" configuration, two "hollow" configurations (fcc and hcp) and one "bridge" point. It is reasonable to assume that the most stable configuration possess also some symmetry: for all the configuration taken in exam, the bigger atom (O in H_2O and N in NH_3) is placed on one of these sites. Physisorption energy is defined as the energy difference between the cell with the molecule and the sum of the energies of the slab and the isolated molecule:

$$E_{phys} = E_{slab+mol} - E_{slab} - E_{mol}.$$

The energy of isolated molecule is obtained constructing a cubic supercell of 20 Å and relaxing the structure. Bond angles and bond length can be compared with known values to test if this approach introduces a large error: with a too small supercell the molecule can interact with its replicas, introducing a systematic error. For the water molecule, bond angle and bond length deviate less than 2%. Molecular adsorption of H_2O is tested for a 3 and 4 layer slab, obtaining an adsorption energy of $-0.35/-0.36$ eV with an O-Cu distance of 2.70/2.66 Å, respectively. We have also tested the effectiveness of DFT-D2 Van der Waals correction [4] in treating the attraction between the permanent dipole of the two molecules and the induced dipole on the slab, that is not considered in PBE exchange-correlation functional. Chemisorption energy is defined as follow:

$$E_{chem} = E_{slab+frag} - E_{slab} - E_{mol},$$

where the fragments are in their most stable (individual) configuration. Dissociation energy E_{diss} is defined as the difference between physisorbtion and chemisorption energy:

$$E_{diss} = E_{phys} - E_{chem} = E_{slab+mol} - E_{slab+frag}.$$

This quantity tells us if the breaking of molecular bonds become favourable or not in presence of the Cu(111) surface. The results for the molecule or the individual fragment are the following:

- H_2O: oxygen prefers the top site over the others, with an energy gain of about -0.1 eV. During the relaxation, the molecule tends to align horizontally with respect to the Cu layers.

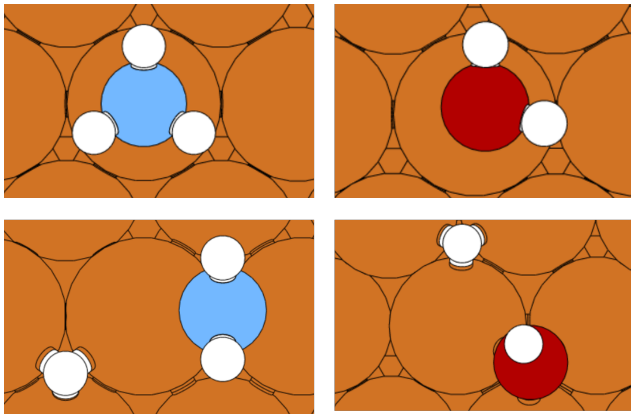


Figure 3: Most stable configuration for physisorbed (upper) and chemisorbed (lower) state of NH_3 (left) and H_2O (right)

- OH: this species is stable only on hollow sites, with fcc slightly preferred over hcp (-0.04 eV). The alignment is vertical
- H: stable on hollow sites, fcc is preferred (-0.05 eV)
- NH_3: top position of N atom is preferred over hollows (-0.3 eV). The configuration is similar to that of the isolated molecule, with H atoms oriented far from the surface.
- NH_2: horizontal position is highly unfavourable, bridge position is preferred over hollows, top is unstable. Hydrogens point to hollow sites.

After this analysis, we have decided to set the physisorbed and chemisorbed states as in Fig. (3). Dissociation energies depend greatly on the initial and final configurations: with a different choice of E_{cut} and n_k these may change. E_{diss} varies also a bit switching on DFT-D2 corrections, although these two methods make the same predictions on the most stable configurations. This predictions agree with literature [9]. Results are in Tab. (2), with some other comparisons of the ammonia configurations.

Energies (eV)	PBE	DFT-D2
diss (H_2O)	0.40	0.26
diss (NH_3)	0.41	0.52
top-hcp (NH_3)	-0.26	-0.33
bdg-fcc (NH_2)	-0.21	-0.23
fcc-hcp (H)	-0.01	-0.01

Table 2: Comparison between dissociation energies of H_2O and NH_3 obtained with normal PBE and PBE+DFT-D2. Energy differences between some high-symmetry configurations of NH_3 and its radicals are displayed as further benchmark.

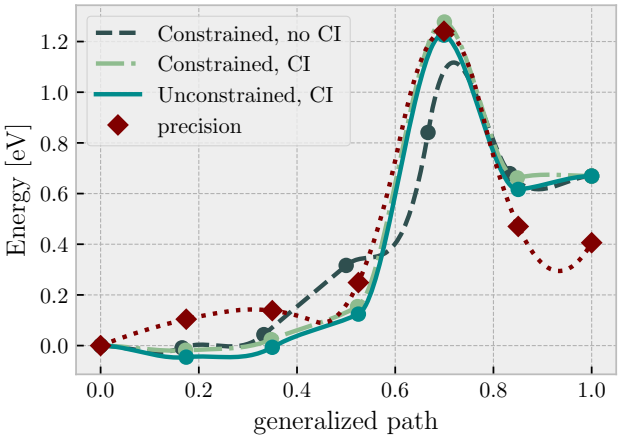


Figure 4: Results of NEB calculation, in constrained paths only the upper Cu layer is free to move. First point to the left is associated to physisorbed H_2O, last point to the dissociated molecule. The most reliable points are precision one (red).

3.4 Reaction path

Reaction path is the path (in the configuration space) followed by the system during the transition from initial to final state. It is reasonable to assume that this path would be the Minimum Energy Path (MEP) among all the possible. Once this path is found, we can characterize entirely the reaction, knowing the energy barrier height in both directions. We have characterized H_2O dissociation in this way, using the Nudged Elastic Band (NEB) algorithm, running with 7 images. Results are shown in Fig. (4).

We have performed three fast NEB calculations, with $E_{cut} = 25$ Ry and a $2 \times 2 \times 1$ k-grid: in the first we have constrained the two lower layers; in the second we have climbed the image near the energy barrier; in the third we have left all atoms free to move. This NEBs are needed only to find the atomic coordinates of the high-

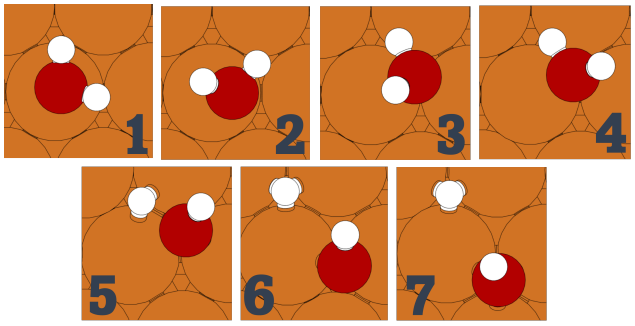


Figure 5: Pictorial representation of the found MEP path found in Fig. (4) *(unconstrained, CI).*

est energy image – through the Climbing the Image (CI) method – since with this cut-off all the results are only qualitative. As a consequence, dissociation energy in this calculation is found to be almost doubled. Assuming that with this low precision calculations we have found the right highest energy (saddle) point among the MEP, we have performed high precision scf ($E_{cut} = 30$ Ry with a $3 \times 3 \times 1$ k-grid) of these images and find a reliable barrier height, resulting in 1.24 eV towards dissociation (left to right) and 0.84 eV on the other direction. These results are in good agreement with literature [10] (1.28 eV) [11] (1.17 eV).

3.5 Interface reaction

H_2O dissociation is also studied in different thermodynamical environment: we have constructed an interface – replicating the slab in a specular way – and analysed the behaviour of the water molecule in presence of various loads, applied to the outer layers of the slabs. As shown in Fig. (6), water does not split even with the greatest load (21 Gpa). Results on energy difference and interfacial spacing are summarized in Tab. (3).

Load (GPa)	ΔE (eV)	Int. spacing (Å)
0	0	5.3
7	0.94	4.1
14	2.03	3.4
21	3.21	3.1

Table 3: Energy increment ΔE and mean interfacial spacing with different applied loads.

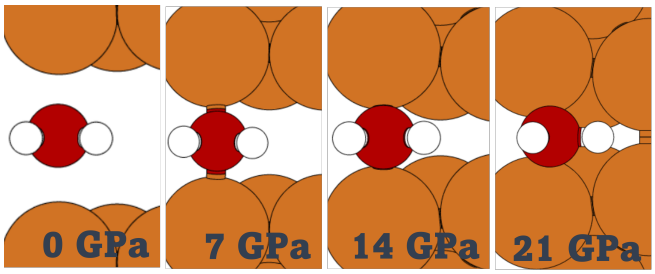


Figure 6: Lateral view of Cu-H_2O-Cu interface, under different loads.

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